Department of Meteorology

Charge-Enhanced Nucleation in WRF's Spectral Bin Microphysics

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Introduction

Cloud and fog droplets naturally become electrically charged. Charge modifies the Köhler curve, changing activation thresholds. Earlier droplet activation could influence cloud growth, evaporation, reflectivity, and persistence.

Köhler theory

- The charged form of the Köhler equation is given by:

$$S - 1 = \frac{A}{r} - \frac{B}{r^3} - \frac{CQ^2}{r^4}$$

- A charged droplet is more difficult to evaporate because reducing its radius would require additional electrostatic energy.

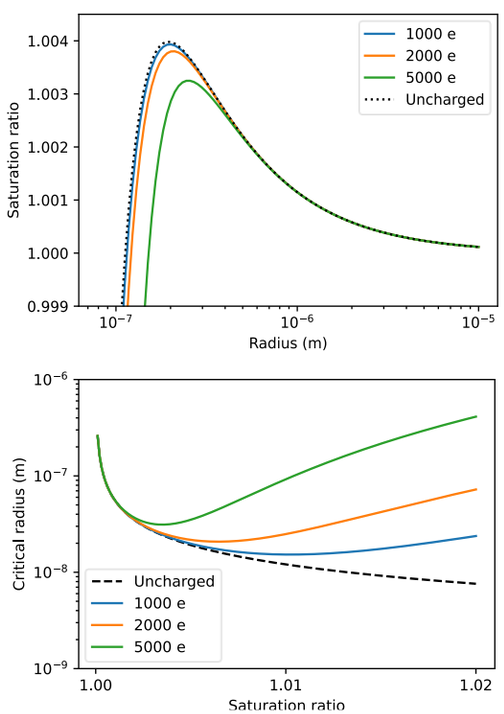


Figure 1. Köhler curves for a range of elementary charges per drop (above). The equivalent implementation in WRF's SBM (below).

Charged nucleation

We incorporate the effect of electric charge on droplet nucleation into the Fast Spectral Bin Microphysics scheme of the Weather Research and Forecasting (WRF) Model.

A charge perturbation term is applied to the pre-existing critical radius term:

$$\delta = \frac{CQ^2(S^* - 1)}{AB}$$

$$r_c = (1 + \delta)\, r_0$$

With **Q = 1000 elementary charges**, the critical activation radius required to nucleate a CCN increases by around 1%.

Initial Results

- The effects of charge enhanced nucleation are nonlinear and have both positive and negative impacts on size.
- Some thin areas of cloud nucleate and persist gently where they would not otherwise.
- Overall, approximately increasing the mean droplet size by 1% for this case.
- Effects of this overall on cloud lifetime and reflectivity are yet to be explored.

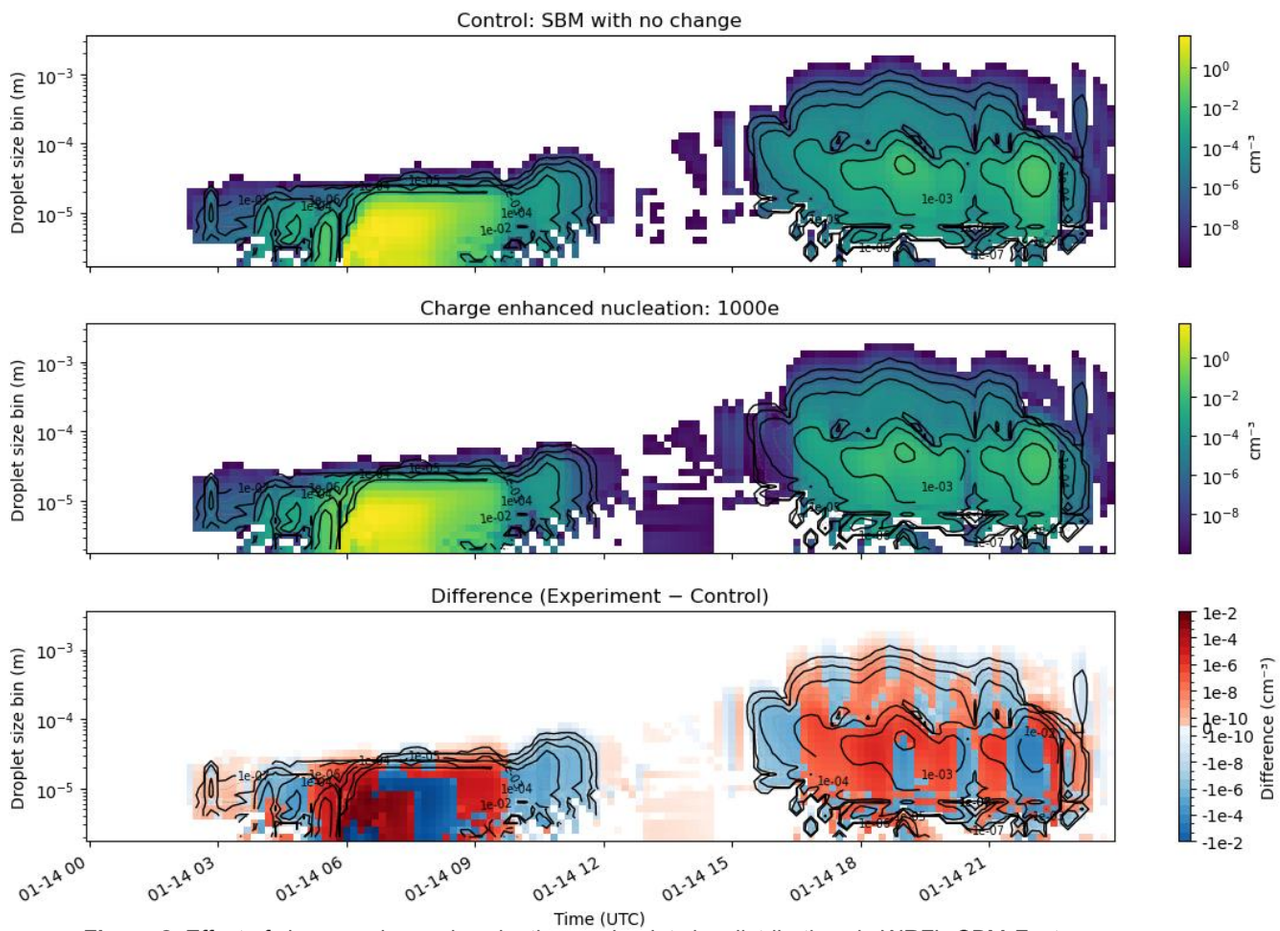


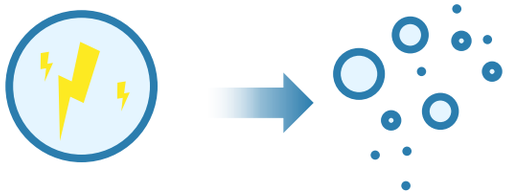
Figure 2. Effect of charge-enhanced nucleation on droplet size distributions in WRF's SBM-Fast microphysics for an example day (14th January 2026) in Reading, UK.
Charge used is 1000 electrons per drop: note that this charge is very high for natural clouds.

Future work

We plan to apply and model more impacts of charge in WRF, including:

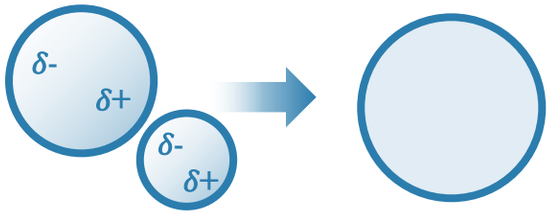
Rayleigh explosions

In highly charged droplets, electrostatic repulsion between like charges on the droplet's surface overcomes the surface tension that holds the droplet together:



Enhanced collision coalescence

Charged droplets induce dipoles in one another, enhancing attraction between droplets:



We also plan to implement more realistic distributions of charge.

About

- Previous work includes laboratory and field experiments on the aerodynamics and orientations of ice crystals.
- I enjoy investigating seemingly small cloud microphysical mechanisms that have large scale radiative impacts.
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